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Degrees, arcminutes, and arcseconds: cutting a circle into small pieces

Astronomers and navigators measure the sky with angles, and angles come in three sizes: degrees, arcminutes, and arcseconds. Each size is sixty times smaller than the one before.

1. A full turn is 360 degrees

Turn all the way around and you have turned 360 degrees, written 360°. A quarter turn, from facing north to facing east, is 90°.

Hold your arm straight out and look at your pinky. It covers about 1° of sky. The full Moon is about half a degree, so your pinky can hide it.

2. Each degree splits into 60 arcminutes

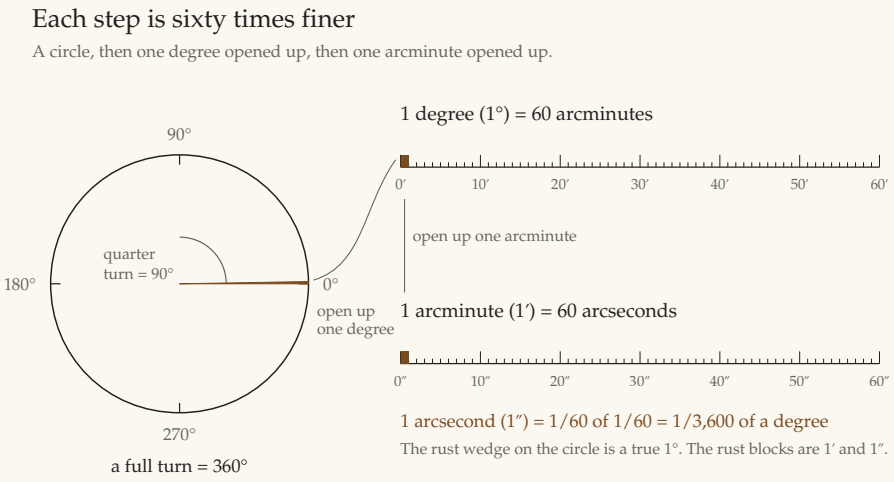
One degree is cut into 60 equal pieces. Each piece is one arcminute, written 1′.

The word "arc" means these are pieces of an angle, not pieces of an hour. An arcminute has nothing to do with a minute on the clock. The two only meet when you turn time into longitude, which is a separate lesson: Time to longitude.

3. Each arcminute splits into 60 arcseconds

Cut one arcminute into 60 pieces again. Each piece is one arcsecond, written 1″.

So one degree holds 60 × 60 = 3,600 arcseconds, and **1″ = 1/3,600 of a degree**.



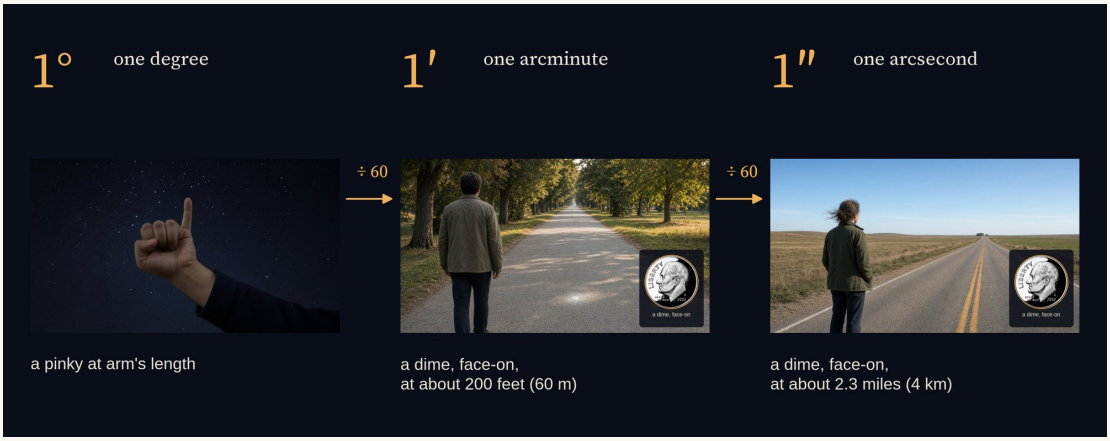
A full turn is 360°. One degree opens up into 60 arcminutes. One arcminute opens up into 60 arcseconds.

Diagram drawn for this lesson. The rust wedge on the circle is a true 1°.

4. What they look like: a dime, seen face-on

Picture a dime, looking straight at its face, its full 18 mm width, not its thin edge.

- Your pinky at arm's length: about **1°**.
- The dime, face-on, at about **200 feet** (60 meters): about **1′**.
- The same dime, face-on, at about **2.3 miles** (4 kilometers): about **1″**.



A pinky at arm's length is 1°; a dime, seen face-on, at about 200 feet is 1′; the same dime, face-on, at about 2.3 miles is 1″

The same illustration as the angle ladder lesson. Dime inset: a 2002 Roosevelt dime, public domain, via Wikimedia Commons.

5. On the ground: one arcminute is one nautical mile

Walk north and the angle of your latitude climbs. One degree of latitude is 60 arcminutes, and each arcminute of latitude is one nautical mile. So one degree of latitude is 60 nautical miles. That is how the nautical mile was born. Today it is fixed at exactly 1,852 meters.

A worked example: one arcsecond of latitude is 1/60 of a nautical mile, 1,852 ÷ 60 ≈ 31 meters, or roughly 100 feet. Reading a map to the nearest arcsecond places you within about a house lot.

6. Why the tiny unit matters

When Earth moves from one side of its orbit to the other, a nearby star seems to shift very slightly against faraway stars. That shift is called parallax. A star whose parallax is one arcsecond sits at a distance of one **parsec**, about 3.26 light-years. Every real star is farther than that, so star shifts are all smaller than an arcsecond.

The rest of the climb, down to what the Gaia space telescope can measure, is in The angle ladder: from a pinky to Gaia and Gaia: ESA's map of the Galaxy.

Optional note: dividing by 60 comes from the ancient Babylonians, who counted in sixties. The same habit gives us 60 minutes in an hour.

Key facts

- A full turn is 360°. A quarter turn is 90°. A pinky at arm's length covers about 1°.
- 1° = 60 arcminutes (60′). 1′ = 60 arcseconds (60″). So 1″ = 1/3,600 of a degree.
- "Arc" means a piece of an angle, not a piece of clock time.
- A dime, seen face-on, covers about 1′ at 200 feet (60 m) and about 1″ at 2.3 miles (4 km).
- One arcminute of latitude is one nautical mile, so 1° of latitude is 60 nautical miles; the nautical mile is now fixed at 1,852 meters.
- One arcsecond of latitude is about 31 meters, roughly 100 feet.
- A parallax of 1″ means a distance of one parsec, about 3.26 light-years.